JeshlaAutoAI Workstream Roadmap

Progress and completion checklist | 17 October 2026

Core AI agent roadmap. Workstream 1 is Y; remaining gaps in 2 and 3 are P and parked. Workstream 4 is active. Baseline: 241 offline tests passed. Core release and framework expansion remain pending.

Learning phases remain Phase 1 Y, Phase 2 implementation Y and Phase 3 I. Workstreams track completion without restarting learning.

Completion flow Foundation Y → Conversations P → Memory P → Knowledge I → Reasoning P → Actions P → Reliability P → Release X → Framework P

Status key

Y Verified scope X Not completed or not started
I In progress P Pending verification or parked

Workstream overview

Status	Workstream	Acceptance focus
Y	1 Foundation	Three configured providers pass scoped direct and tool/result acceptance; Ollama compatibility retained.
P	2 Conversations and context	Working chats, shared memory and downloads verified. Summary quality and full live summary isolation parked.
P	3 Long term memory	Facts, provenance, revisions, archive search and suggestions verified. Semantic and broad conflict quality parked.
I	4 Knowledge and evidence	Reference guards, page reader and keyword ranking implemented. Mixed-source and live entailment checks next.
P	5 Reasoning and task completion	Basic loop works; complex planning and recovery acceptance pending.
P	6 Controlled actions	Console approval coverage exists; browser/action replay acceptance pending.
P	7 Reliability	Errors, budgets, timings and browser logs implemented; broader recovery remains pending.
X	8 Core agent release	Whole core acceptance, clean setup and backup/recovery pending.
P	9 Automation framework integration	Deferred until core release acceptance and explicit user decision.

Workstream 1 Foundation

Status Y. Direct answer and real tool/result cycle verified for the three selected models; safe configuration errors shown in both interfaces.

Live acceptance passed for OpenRouter nvidia/nemotron-3-super-120b-a12b:free, Groq openai/gpt-oss-20b and Ollama qwen2.5-coder:7b. Ollama uses a narrow registered-JSON compatibility path; existing tool validation and action approvals remain.

Status	Completion checklist
Y	Groq connection works with the configured open-weight model (live execution).
Y	Native model/tool/result loop works with public web search (live execution).
Y	Ollama/OpenRouter/Groq adapters exist; alternate native adapters have offline transport tests.
Y	The three selected provider/model combinations pass live direct-answer and real tool/result acceptance: OpenRouter, Groq and Ollama. Ollama JSON-text proposals are narrowly normalized to the shared tool-call format; caller validation and approvals remain. See foundation-acceptance.md.
Y	Actionable, secret-safe provider configuration errors in console and browser: missing hosted keys/model settings, invalid numeric settings and unknown provider; shared presentation and headless UI regressions pass (181 offline tests).

Workstream 2 Conversations and context

Status P. Console recall/switching and browser create/chat/reopen/memory/download are verified for observed scope. Actual-history summary accuracy and full live summary isolation remain parked.

Status	Completion checklist
Y	Create, list and reopen named chats (user execution and offline tests).
Y	Case-insensitive startup Y/N and 80-character titles work (offline tests and user execution).
Y	Editable console multiline input works without END (user execution).
Y	Save full newly successful turns in a transcript archive (offline persistence tests).
Y	Streamlit supports chat creation, multiline submission and memory viewing in headless tests; real startup passed.
Y	Summary failures pause retries, including across restarts (user output and offline checks).
Y	Implement and validate the registered summary path: 189 offline tests pass; synthetic long-chat summaries pass on Groq and OpenRouter. The exact cause of the historical HTTP 400 is unconfirmed.
Y	Verify summary persistence, restart, chat switching and complete transcript preservation with real SQLite storage and mocked models.

Workstream 2 continued

Status	Completion checklist
P	Actual-history summary accuracy: Groq format rejection and exhausted quota block live diagnosis; Ollama completed 6/7 grounding checks but failed source attribution. Resume later; see workstream-2-summaries.md.
Y	User verified console restart recall and session switching; browser screenshots, separate transcript exports and live event logs verify sampled chat separation.
P	Complete live summary-content isolation across restart/switch after actual-history summarization is reliable. Existing SQLite/model-mocked summary isolation tests pass.
Y	Browser create/chat/reopen/shared-memory/download acceptance passed for the observed scope: separate browser/default transcript exports and user screenshots confirm distinct chats and shared saved facts. Page-jump usability remains separate.

Parked gaps: Groq actual-history summary-format reliability and Ollama source attribution. Do not spend further quota on these until resumed by user decision.

Workstream 3 Long term memory

Status P. Provenance, timestamps, explicit revision history, lexical archive search and controlled suggestions are verified. Semantic relevance and broad contradiction accuracy remain parked after the bounded local retry.

Status	Completion checklist
Y	Save, update, show and forget explicit facts (offline tests; save/show/forget also user verified).
Y	Deleted seed facts are not automatically reinserted (offline regression).
Y	Memory persists across store reopen and is scoped by user/namespace (offline tests).
Y	Saved facts reach normal LLM conversation (user execution).
Y	Optional local-only MiniLM semantic retrieval is integrated with scoped SQLite facts and search_memory; lexical default preserved.
P	Real semantic relevance calibration: bounded local retry loaded the model but returned no paraphrase match. Mocked scope/update/delete tests pass. No downloads or API calls.
Y	Bounded lexical archived-conversation search is connected to the shared agent, scoped to user/memory-space and current or explicitly selected chats, with chat/turn/time references. 210 offline tests and an actual local archive lookup pass; hosted tool-choice acceptance remains P.
Y	Fact provenance, existing timestamps and verification labels are stored and exposed in memory context/commands. Backward-compatible migration preserves all 5 existing database rows; 203 offline tests pass. See workstream-3-memory.md.

Workstream 3 continued

Status	Completion checklist
Y	Explicit same-key corrections preserve scoped revision history; normal retrieval uses current values only. MEMORY HISTORY key reviews revisions; FORGET deletes associated history. 206 offline tests pass.
Y	Advisory related-fact conflict candidates and configurable storage-age review are implemented; no automatic merge, deletion or truth claims.
P	Broader semantic contradiction quality across keys/projects/timeframes: real HTML/PDF advisory smoke case passed, but general accuracy remains unverified.
Y	Scoped suggestions require explicit ID-bound accept/reject; no automatic saved facts or existing-key overwrite. 214 offline tests and temporary command integration pass; live model suggestion selection remains unverified.

Workstream 4 Knowledge and evidence

Status I and active. Web/document reference guards and bounded public-page retrieval are implemented. Keyword retrieval passes 8 of 8 audited source/keyword cases, or 6 of 8 unchanged original labels; two page labels were verified against the PDF. Next: mixed-source grounding and live citation acceptance. Mixed-source offline integration passed. The live Groq fixture run selected all four tools; final answer generation hit quota. Live final-answer acceptance remains P.

Status	Completion checklist
Y	Implement document ingestion, metadata and recursive chunking in rag/ingest.py; artifacts are present.
Y	Implement FAISS/BM25 hybrid retrieval and BGE reranking in rag/retrieval.py and rag/query.py.
Y	Implement query rewriting and PDF answer export in rag/query.py.
Y	Implement CSV/Excel search, header detection and dataset routing in the rag modules.
Y	Implement RAG tool-calling/web agents and evaluation variants; saved reports document prior runs.
P	Consolidate older provider-specific RAG entry points and working-directory-sensitive imports/paths with the current application.
Y	Public web search returns real snippets and links (live execution).
Y	A response can combine saved project facts with web results (user execution).

Workstream 4 continued

Status	Completion checklist
Y	Web reference traceability guard rejects unseen URLs and invented internal markers, permits one correction, and safely fails empty required searches. 225 offline tests pass.
I	Validate factual entailment, publisher/source quality and local/document citations; a registered URL alone does not prove a claim.
Y	Bounded static public-page reader is connected to search results with DNS pinning/public-address validation, redirect checks, excerpt limits and final-URL citation registration. 232 offline tests pass.
P	Live page compatibility, hosted source selection and claim-to-page entailment acceptance; hard end-to-end network deadline remains a limitation.
Y	Re-evaluated existing keyword indexes locally: baseline 1/8 expected source/location/keyword cases; report document-keyword-baseline.json.
Y	Normalized/source-aware keyword ranking and bounded same-page context pass 8/8 audited source/location/keyword cases (6/8 original labels; two page errors independently checked against PDF). 240 offline tests pass.
P	Broader document relevance, answer entailment, fresh-source coverage and hybrid quality remain unverified beyond this small keyword benchmark.
Y	Structured indexed source IDs, bounded excerpts and document-citation validation are integrated while legacy text/hybrid entry points remain compatible. 237 offline tests pass.

Workstream 4 continued

Status	Completion checklist
P	Integrate and test structured CSV/Excel retrieval in the active conversation loop.
X	Pass a mixed memory/document/source/web task with traceable, accurate evidence.

Workstream 5 Reasoning and task completion

Status P. Complete multi-step tasks, recover from failed tools and report complete, partial or blocked outcomes accurately.

Status	Completion checklist
Y	Read-only loop validates tool arguments and limits rounds/calls (offline tests).
Y	Unclear ordinary input receives clarification (user execution).
P	Evaluate dynamic clarification on a representative question set.
P	Evaluate multi-step planning, failed-tool recovery and re-planning.
P	Verify completed/partial/blocked outcomes reflect actual evidence.

Workstream 6 Controlled actions

Status P. Bind approval to the exact action; prevent restart or cancellation from replaying side effects.

Status	Completion checklist
Y	Console exact-action approval has regression coverage.
Y	Browser local-action policy does not fall back to hidden console approval (implementation and scoped-policy test).
P	Add and verify exact-action browser approvals, or explicitly accept the console-only limitation for the release.
P	Verify restart, cancellation and uncertain outcomes cannot replay completed side effects.
P	Validate action scope and permissions across all tools supported in the core release.

Workstream 7 Reliability

Status P. Handle provider, budget, concurrency and recovery failures with observable evidence and repeatable tests.

Status	Completion checklist
Y	Timings and bounded tool execution are implemented and exercised.
Y	Same-chat concurrency locks block overlapping turns and release normally (offline tests).
P	Exercise provider/network/quota/truncation failures through each supported interface.
X	Document and test stale-lock recovery, archive failures and interrupted checkpoints.
P	Complete context budgeting and relevant-evidence selection for larger tasks.
Y	Guarded broad-suite isolation and old Jira fixtures repaired; latest regression baseline is 241 passing tests. This is not full release acceptance.

Budget note: the project input guard is 32000 characters. Provider preflight estimates tokens; it does not guarantee exact token fit or remove account quotas.

Workstream 8 Core agent release

Status X. Pass the whole core checklist, clean setup and backup/recovery; agree the release scope.

Status	Completion checklist
X	Pass the full core-agent acceptance checklist in agentic-ai-roadmap.md.
X	Verify clean installation using the documented dependencies and configuration.
X	Update outdated root README and installation instructions.
X	Document and verify backup/recovery of facts, chats, summaries and checkpoints.
X	Review remaining limitations and explicitly agree on the release scope.
X	Mark the general AI-agent milestone complete.

Do not mark the whole agent complete until the core acceptance checklist, setup and recovery checks pass and release limitations are agreed.

Workstream 9 Automation framework integration

Status P. Start only after core release acceptance and the user chooses framework work.

Status	Completion checklist
P	Begin only after the core-agent release gate passes and the user chooses to proceed.
P	Create a separate framework-integration roadmap at that time.

Existing Jira and QA modules are preserved. New framework development remains parked until core release acceptance.

Evidence and document maintenance

Updated snapshot of agentic-ai-roadmap.md through the keyword ranking milestone and 241-test mixed-source offline check. Detailed evidence: workstream-2-summaries.md, workstream-3-memory.md, workstream-4-grounding.md and browser-logging.md. Small benchmark coverage does not prove universal answer accuracy.

Update both the roadmap and this Word snapshot after each completed workstream. Keep the Y X I P codes, state the tested scope and preserve pending release gates.

Restored roadmap explanations and diagrams

The current workstream checklists above remain the status authority. These restored sections preserve the learning phases, architecture, completion diagrams and release criteria. Historical milestones retain their original dates and test counts.

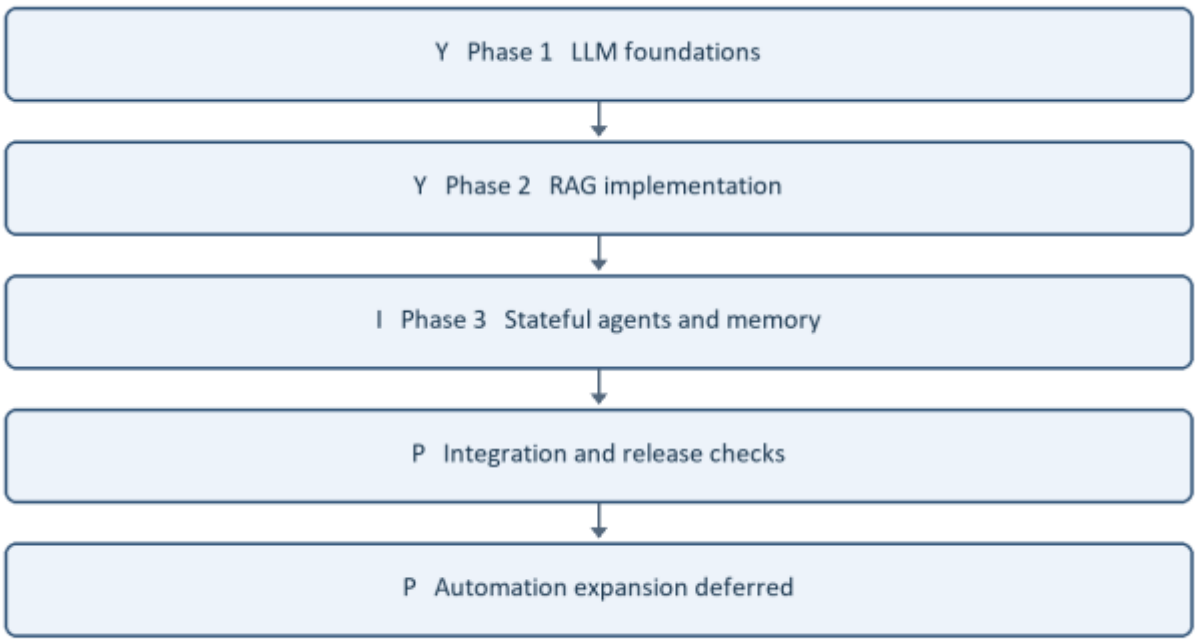
Your actual development position

You are not still in Phase 1. Your current original learning phase is Phase 3: memory, stateful agents and integration hardening.

Original development phase	Build status	Readiness
Phase 1 — LLM foundations and basic agents	Y	Foundation implemented and real model use demonstrated; some release checks remain
Phase 2 — RAG and knowledge implementation	Y	Ingestion, hybrid retrieval, reranking, structured search and evaluation exist; quality/runtime integration is P
Phase 3 — State, persistent memory and conversational agents	I	Core components work; summaries, advanced recall and broader acceptance remain P

The numbered sections later in this document are completion workstreams, not a claim that you must restart Phase 1. A built module can have pending integration or acceptance without losing credit for its implementation.

The full-project audit read/parsed 94 Python files, with no syntax errors. See project-progress-audit.md (project-progress-audit.md) for folder evidence, RAG details, prototype distinctions and the current full-suite test limitations.



Target and scope

Build JeshlaAutoAI as a reliable, local-first AI agent with a configurable hosted or local model, conversation, persistent memory, evidence retrieval, validated tools, and user-controlled actions.

AI agent is the application we are building. Agentic AI describes the approach: the model selects tools, observes results, adjusts its next step, and answers or asks for clarification. Agentic by itself is an adjective, not a separate product target. Recommended project description: JeshlaAutoAI — an AI agent built with agentic AI capabilities.

Automation-framework development is deferred until the core-agent acceptance gates below pass. Preserve existing Jira/QA modules, but do not make new framework integration the next milestone. Completion means the agreed core scope is verified, not that every possible AI capability exists or that the project matches proprietary products.

Memory development details

- Separate chat history, session summaries and reusable facts; never use a summary as approval.
- Select relevant memories instead of injecting every fact into each prompt.
- Keep source, timestamp and scope for facts; distinguish user assertions from tool evidence.
- Search archived conversations only within the intended scope and cite the originating chat/turn.
- Propose useful new memories or corrections; define explicit user control before automatic storage.
- Handle conflicts, stale facts, deletion and optional expiry. Document that deleting a fact does not erase archived messages or provider records.
- Start with a measured retrieval baseline; add embeddings/semantic retrieval only when it improves the project's evaluation set without reintroducing the previous local inference bottleneck.

Implementation principles

- Do not hardcode project names, framework choices, answers or fixed clarification scripts in model reasoning.
- Store editable project facts in memory/configuration. Keep credentials outside source and logs.
- Use deterministic code for command parsing, argument validation, permission checks, isolation, budgets and storage.
- Prefer documented configurable defaults for limits; not every internal invariant needs a user-facing option.
- Use model reasoning for interpreting ordinary requests, choosing read tools, synthesizing evidence and asking focused questions.
- Add tools with schemas and clear failure behavior. Native model tool calls do not themselves authorize execution.
- Keep a shared backend for console and Streamlit. UI changes should not change approval rules.
- Use the simplest architecture that passes the required tasks; multiple agents, autonomous background work and additional providers are optional future capabilities, not prerequisites.

Core-agent acceptance checklist

Release gates receive Y only when the complete acceptance scenario passes. A working component alone does not complete a broad gate. P means related work exists but full acceptance is pending.

- P — General conversation and clarification use supplied context without inventing missing facts.
- P — Named chats survive restart and have isolated history/summary; shared memory is scoped deliberately.
- P — Memory save/update/forget and relevant search pass both ordinary and conflicting-fact cases.

- P — Long chats preserve full newly archived transcripts; summaries carry key details without becoming instructions or permissions.
- P — Local source, indexed documents, structured data and public web searches work independently and together.
- P — Citations correspond to retrieved evidence; current facts are not claimed after search failure.
- P — Multi-step tasks stop at a clear result or ask for essential missing details.
- P — Tool arguments, source paths and action permissions are validated in code.
- P — Write/execute approvals work in both supported interfaces, or a documented interface limitation is explicitly accepted for the release.
- P — Failed/uncertain actions are not silently retried; restart and cancellation do not duplicate side effects.
- P — Provider, network, quota and context-limit failures are understandable and recoverable.
- P — Repeatable setup, data backup/recovery and the acceptance/regression suite pass in the supported environment.

Immediate next milestone

Workstream 4 — Reliable knowledge and search is now the active milestone. Verify source citations and retrieved evidence, then current indexed-document quality and mixed-source answers. Workstreams 2 and 3 retain verified Y items but their remaining summary/semantic/conflict acceptance gaps are P—parked by user decision. Do not retry those models now. Automation-framework expansion remains deferred until core release acceptance.

Evidence labels

- Implemented: code exists; does not imply full live acceptance.
- Live verified: supported by an observed real execution.
- Pending: not implemented or not yet verified sufficiently.
- Deferred: deliberately outside the current development scope.

Existing root README installation/END/history notes predate several changes. Reconcile documentation during release preparation; prefer newer topic guides until then. Existing Jira uncertainty remains a tracked framework-integration issue, not something to bypass or delete while focusing on the general agent.

AI agent versus agentic AI

These terms overlap, and terminology varies. They do not describe two mutually exclusive technologies.

Term	Meaning	Example in JeshlaAutoAI
AI agent	A concrete software system that uses a model to pursue a task through available tools and context	Your application reads saved facts, chooses web search, receives results and answers
Agentic AI	An approach or capability in which a system chooses and adapts its actions toward a goal within defined boundaries	The agent notices missing evidence, searches another source, checks the result and decides whether to answer or ask you
Agentic	An adjective describing that behavior	An agentic assistant or agentic workflow
Fixed workflow	A predefined sequence of steps and conditions; useful when the process should be predictable	Command parsing, exact-action approval and database validation

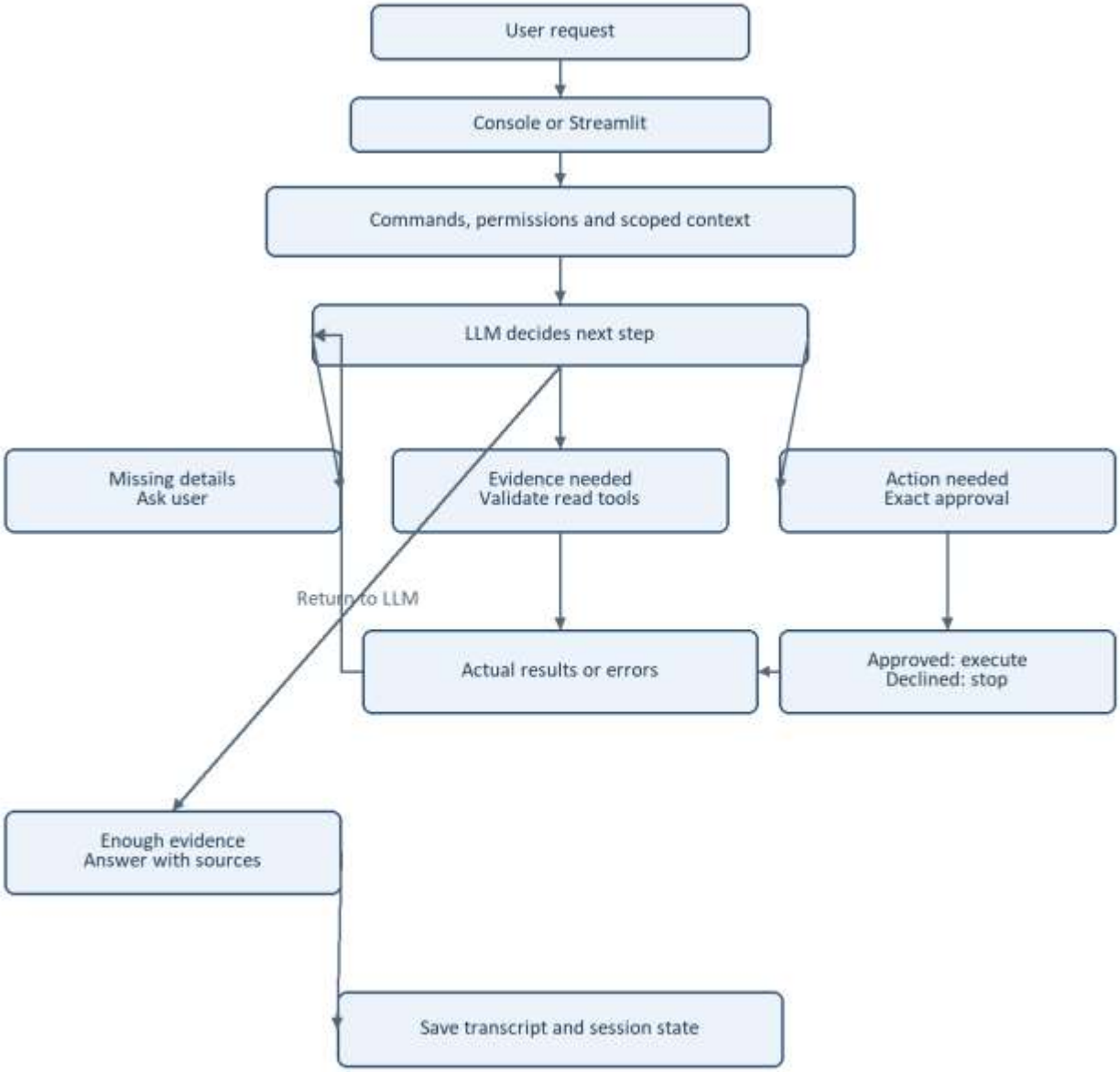
An AI agent can be more or less agentic. Agentic behavior does not require multiple agents, unlimited autonomy, a particular model, or automatic self-training. Fixed workflows can form part of a reliable agentic system.

Which is better for this project?

Build one reliable AI agent with bounded agentic behavior. Let the model interpret requests, select read tools, evaluate results and ask for missing details. Keep permissions, validation, storage, command handling and execution limits deterministic. More autonomy is useful only when its results can be checked; it is not automatically better.

For example, answering a saved fact needs no elaborate plan. Research across several sources benefits from adaptive tool use. A write or command still needs the applicable approval regardless of how the model planned it.

Architecture diagram

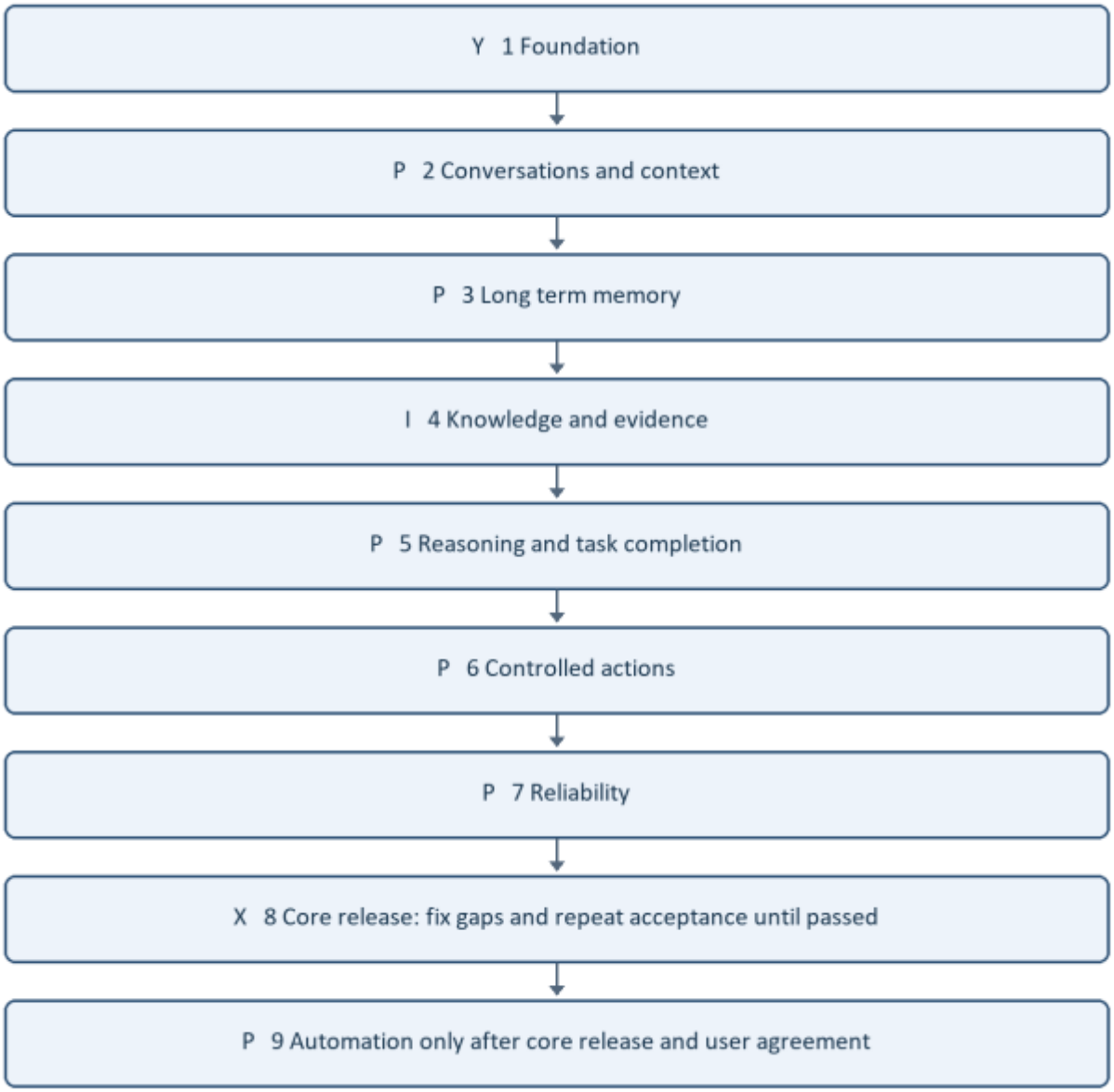


Tool results feed back into the model for another decision. Clarification returns to the user; declined approval stops the action. This is the intended flow, not a claim that all release gates have passed.

This diagram is the intended architecture. The read-only LLM/tool loop is working. Browser action approval, advanced memory retrieval and several reliability gates remain incomplete; the diagram does not claim every box is fully implemented.

Roadmap and completion flow

Status codes: Y = completed, X = not completed/not started, I = in progress, P = worked on but pending/parked. Deferred framework work uses P as a deliberate scheduling status, not a claim that it is implemented.



The recommended target is therefore not AI agent *or* agentic AI: JeshlaAutoAI is the AI agent; agentic AI is the behavior and design approach it uses.

Free and open-source stack priorities

The workbook comparison and project-specific P1/P2/P3 checklist are saved in free-open-source-module-audit.md (free-open-source-module-audit.md). These priorities complement the Y/X/I/P completion statuses above.

- P1: retain the working stack; repair dependency reproducibility, isolated tests, relevant memory/structured retrieval, citations, summary recovery and acceptance evaluation.
- P2: add tracing, evaluation tooling or improved parsing only for a measured gap.

- P3: park alternative orchestration/vector frameworks, fine-tuning, multimedia and automation-framework expansion until the agreed core is accepted.

Free API access is quota-limited and separate from software/model licensing. Installed alternatives do not imply active use or completed integration.

Historical milestone record

Older active-workstream descriptions and test totals below are historical. The current verified baseline is 241 offline tests; Workstream 4 is active. Mixed-source final-answer acceptance remains pending after a Groq quota rejection.

Offline test baseline completed - 17 October 2026

Y - scoped completion: repaired stale test fixtures and verified 120 tests passing in 3.704 seconds using the offline guard. This supersedes the earlier 116-test/six-error baseline for these fixtures; it does not establish live API, RAG answer-quality or production acceptance.

Run from the project root:

```
python tests/run_offline.py
```

Changes:

- tests/test_jira_creation.py: enable creation only within fake-gateway test fixtures; add a regression proving disabled creation blocks writes.
- tests/test_prompt_workflow.py: mock proposal discussion instead of calling the real model; explicitly enable the fake approval workflow in its test scope.
- tests/run_offline.py: guard discovery and execution against external sockets, unmocked requests HTTP calls and browser launches. Loopback sockets remain for asyncio/framework internals. This is a process-level test guard, not a subprocess sandbox.
- tests/test_offline_runner.py: verify socket/HTTP/browser blocking and loopback classification.

The verified run used the project's virtual environment and required Windows file permissions. Real Jira permission gates, provider settings and runtime agent behavior were not changed. Full agent reliability remains P; next is memory and structured retrieval integration.

Shared table and memory tool integration - 17 October 2026

Y - implementation and offline integration: CSV/XLSX discovery/filtering with source rows and scoped lexical saved-memory search are connected to the shared conversation. 129 guarded offline tests pass. See table-memory-tools.md (table-memory-tools.md) for files, limits and trial prompts. P - broader acceptance: live model behavior, semantic-memory retrieval, complex-table queries and grounding remain pending.

Task outcome logging - 17 October 2026

Y - scoped completion: project reports include task_status (completed/incomplete); timer decorators and the console's outer workflow observe that explicit status. Returned incomplete results log INCOMPLETE instead of SUCCESS. Exceptions still propagate and log FAILED / INTERRUPTED. New turns reset task_status so an earlier incomplete checkpoint does not contaminate the next result. Shared conversation budget/invalid-ID/step-limit exits also mark incomplete.

165 guarded offline tests passed in 28.822 seconds. Added tests/test_task_timing.py and updated runtime reset coverage. Files: agent/timing.py, agent/chat_runtime.py, agent/conversation.py, agent/langgraph/langgraph_agent.py. Runtime tools, approvals, provider selection and saved facts were not changed. This covers explicit task outcomes; route-specific blocked/waiting states without task_status still need separate acceptance coverage. No live API call was required.

The user's previous successful settings.gradle/build.gradle run confirmed PayWrightWorkSpaceGradle correctly and inspected all requested lines. Next priority remains provider/model-aware token budgeting, followed by grounded document workflows and overall core-agent acceptance.

Provider-budget preflight - 17 October 2026

Y - configured/estimated preflight: Ollama request context, configurable hosted context/output caps, model binding and explicit unknown-limit diagnostics are implemented. Public OpenRouter model metadata was verified and stored for the selected model. 174 offline tests pass. See model-budgeting.md (model-budgeting.md). P - exact tokenizer/model discovery, quota scheduling, metadata refresh and adaptive compaction remain. This supersedes the earlier entirely-pending provider-budget item without claiming full token accuracy.

Workstream 2 summary reliability started - 17 October 2026

I overall. Registered, data-only summary generation replaces the old JSON-only client path. Validation, one bounded format correction and category-safe failures preserve context/archives. 189 offline tests pass, including real SQLite summary restart/session isolation with model mocks. Live synthetic long-chat summaries pass on Groq (2.11s) and OpenRouter (11.11s), retaining a correction and unanswered question. Full user-chat/live-browser acceptance remains pending; Ollama summary generation was not live-tested. See workstream-2-summaries.md (workstream-2-summaries.md). The Word workstream snapshot is docs/agentic-ai-roadmap.docx.

Workstream 2 status review - 17 October 2026

I — In progress overall. Summary implementation and scoped persistence checks are Y, supported by 189 offline tests and live synthetic Groq/OpenRouter summary reports. Actual user-history quality, full live restart/switch verification and complete browser create/chat/reopen/memory/download acceptance are P. Mark the overall workstream Y only after those remaining acceptance checks pass. See workstream-2-summaries.md (workstream-2-summaries.md).

Workstream 2 item 1 grounding fix — 17 October 2026

Y for scoped archive-supported question retention and source-attribution retest: 194 offline tests and seven live synthetic OpenRouter checks passed. Item 1 remains P for broader actual-history accuracy and Ollama context acceptance. See workstream-2-summaries.md (workstream-2-summaries.md).

Workstream 2 sequencing decision — 17 October 2026

At the user's request, park item 1 (P): actual-history summary format/accuracy and Ollama attribution. Do not spend further API quota on this blocker now. Move next to item 2: complete live restart/chat-switch isolation acceptance; then item 3 browser acceptance. Overall Workstream 2 remains I; parking is not completion.

Browser logging — 17 October 2026

Added local privacy-safe rotating browser event logs; 199 guarded offline tests pass. See browser-logging.md (browser-logging.md). Live event inspection is pending after the user reloads the app. Workstream 2 remains I because summary accuracy is parked P and full live summary isolation remains pending.

Current handoff — Workstream 3 active, 17 October 2026

This update supersedes earlier progress snapshots: Workstream 2 is P overall (parked, not completed). Console recall/switching, sampled browser chat separation, shared-memory display, transcript downloads and live

browser event logging are Y for observed scope. Actual-history summary generation/accuracy, Ollama source attribution and full live summary isolation remain P—parked. Latest regression baseline: 199 offline tests passed. Workstream 3 is I as the selected active workstream; provenance/timestamp/verification implementation is the first planned task, not claimed completed. See workstream-2-summaries.md (workstream-2-summaries.md) and browser-logging.md (browser-logging.md).

Workstream 3 first milestone — 17 October 2026

Y for provenance/timestamp/verification metadata, safe legacy migration and current memory integration. Existing facts backed up and preserved. 203 guarded offline tests passed; no API calls. Workstream 3 remains I. Next: correction/conflict policy and revision handling; advanced retrieval and archived-chat search remain pending. See workstream-3-memory.md (workstream-3-memory.md).

Workstream 3 correction milestone — 17 October 2026

Y for atomic same-key correction/revision handling, scope isolation and forget cleanup; 206 offline tests plus temporary command integration passed. Broader semantic conflicts, stale facts, archive search and memory suggestions remain pending. Workstream 3 remains I.

Workstream 3 archive-search milestone — 17 October 2026

Y for scoped lexical archive lookup and shared tool integration. 210 offline tests pass; actual default-chat local lookup confirms source references. No model calls or downloads. Semantic retrieval, broader conflict/staleness policy and controlled suggestions remain pending; Workstream 3 is I.

Workstream 3 suggestions milestone — 17 October 2026

Y for explicit user-controlled proposal/review/accept/reject lifecycle and existing-key conflict refusal. 214 offline tests pass. Workstream 3 remains I: semantic retrieval and broader contradictions/staleness remain pending.

Workstream 3 remaining-P review — 17 October 2026

Both pending areas now have implementation: optional semantic retrieval and advisory conflict/age review. 220 offline tests pass. Real cached-model quality/latency and broader semantic contradiction evaluation remain P. The slow local test was interrupted intentionally; no completion is claimed for those acceptance gaps.

Current handoff — Workstream 4 active, 17 October 2026

Supersedes earlier active-focus snapshots: Workstream 3 is P overall after its one bounded real-model retry failed paraphrase relevance. Local loading succeeded; advisory HTML/PDF candidate detection/no mutation passed. Semantic calibration and broader conflict accuracy remain parked, not completed. Workstream 4 remains I and is now the next active scope. 220 offline tests remain the regression baseline.

Workstream 4 started — 17 October 2026

Y for web citation traceability and failed-required-search protection in the shared conversation loop. 225 offline tests pass; no live API calls. Workstream 4 remains I. Full-page retrieval, local document/source citation validation, indexed retrieval quality and mixed-source acceptance remain pending. See workstream-4-grounding.md (workstream-4-grounding.md).

Workstream 4 source-page milestone — 17 October 2026

Y for public-page reader and search/read/citation offline integration; 232 tests pass. No live network/provider calls. Workstream 4 remains I; local-document citations, real retrieval quality, mixed-source acceptance and live page compatibility are pending. See workstream-4-grounding.md (workstream-4-grounding.md).

Workstream 4 document-source milestone — 17 October 2026

Y for structured indexed-source references and citation guard integration; 237 offline tests pass. P for retrieval quality: actual keyword baseline 1/8, resume pages present but other documents rank ahead. No model calls or downloads. Next: generic tokenization/ranking improvements and repeatable re-evaluation, then mixed-source grounding.

Workstream 4 keyword ranking verified — 17 October 2026

Keyword baseline now 8/8 with audited page labels, versus 6/8 unchanged original labels (two incorrect page-1 labels verified as page 2 in the actual PDF). Initial baseline was 1/8. 240 offline tests pass; no provider quota or embeddings used. Legacy hybrid defaults remain. W4 I: mixed-source and live citation/entailment acceptance next.

Mixed-source grounding in progress — 17 October 2026

241 offline tests pass, including scoped memory/document/web/page integration. Live Groq synthetic-fixture tool selection executed all four tools; final-answer request hit QUOTA_EXHAUSTED HTTP 429. Live combined-answer/citation acceptance remains P; no further retry or silent provider change. Word roadmap snapshot updated. Workstream 4 remains I.

Module usage reference

Restored classifications from the 17 October audit. Usage, optional additions and alternatives are distinguished. Historical acceptance notes do not override the current checklists.

Complete module list: usage first

Usage reflects the inspected application paths, not just installed packages. NOT USED means no integration was identified in the audited project, not proof that the package is absent from the environment. OPTIONAL means add only for a demonstrated need; ALTERNATIVE means it overlaps an existing choice. These usage labels are separate from the Y/X/I/P completion checklist.

Usage / decision	Module	Priority	Purpose or evidence
USED	LangChain	P1	Existing agent/RAG integrations.
NOT USED - ALTERNATIVE	LlamaIndex	P3	Alternative RAG framework; retain current implementation.
NOT USED - OPTIONAL	LiteLLM	P2	Provider gateway; existing adapters already cover three providers.
USED - DEPENDENCY	Transformers (Hugging Face)	P1	Supports local sentence-transformers models.
USED - SUPPORTED FALLBACK	Ollama	P2	Provider implemented and previously run; Groq is currently selected.
USED	sentence-transformers	P1	Local embeddings and reranking.
PARTIAL - ALTERNATIVE	ChromaDB	P3	Installed with stored artifacts; inspected active hybrid path uses FAISS.
USED - DEPENDENCY	PyTorch	P1	Local embedding/reranking model execution.
NOT USED - ALTERNATIVE	Haystack	P3	Alternative pipeline framework.

Reference table continued

Usage / decision	Module	Priority	Purpose or evidence
MIXED - DEPENDENCY / OPTIONAL	Tokenizers / Datasets (HF)	P1 / P3	Tokenization through the model stack; no application dataset-library workflow established.
NOT USED - OPTIONAL	PEFT / TRL (HF)	P3	Fine-tuning is outside current acceptance scope.
INDIRECT / OPTIONAL	llama.cpp	P3	Earlier Ollama logs show its llamacpp runner; no separate direct integration required.
NOT USED - OPTIONAL	Google Gemini SDK	P3	New provider adapter needed before use.

Usage / decision	Module	Priority	Purpose or evidence
USED	FAISS	P1	Vector retrieval in the existing hybrid RAG pipeline.
USED	LangGraph	P1	Agent graph, state and checkpoint flow.
NOT USED - ALTERNATIVE	Gradio	P3	Streamlit already supplies browser UI.
NOT USED - OPTIONAL	DSPy	P3	Prompt/program optimization if later justified.
NOT USED - OPTIONAL	bitsandbytes	P3	Quantization tooling not part of the current app path.

Reference table continued

Usage / decision	Module	Priority	Purpose or evidence
NOT USED - OPTIONAL	Unsloth	P3	Fine-tuning tooling.
NOT USED - ALTERNATIVE	vLLM	P3	Separate model-serving runtime.
NOT USED - ALTERNATIVE	CrewAI	P3	Alternative agent orchestration.
NOT USED - ALTERNATIVE	AutoGen	P3	Alternative agent orchestration.
NOT USED - ALTERNATIVE	smolagents (HF)	P3	Alternative agent framework.
NOT USED - ALTERNATIVE	TensorFlow / Keras	P3	Current model stack uses PyTorch.
USED	Streamlit	P1	Browser chat implemented alongside console; live acceptance remains pending.
NOT USED - OPTIONAL	Diffusers (Hugging Face)	P3	Image/video generation outside text-agent target.
NOT USED - OPTIONAL	ComfyUI	P3	Multimedia workflow UI outside current target.

Reference table continued

Usage / decision	Module	Priority	Purpose or evidence
NOT USED - OPTIONAL	MarkItDown	P2	Consider only for a demonstrated document-loader gap.
NOT USED - OPTIONAL	Docling	P2	Consider for complex layout extraction.
NOT USED - OPTIONAL	Unstructured	P2	Alternative document parsing/chunking.
NOT USED - OPTIONAL	PyMuPDF	P2	Alternative PDF parser; license review required before adoption.
NOT USED AS EMBEDDINGS - ALTERNATIVE	BGE / E5 / Nomic embed models	P2	Current embedding model is MiniLM; BGE reranker below is a different role.

Usage / decision	Module	Priority	Purpose or evidence
NOT USED - ALTERNATIVE	Qdrant (self-hosted)	P3	Storage replacement/extension only if requirements justify it.
NOT USED - ALTERNATIVE	LanceDB	P3	Alternative vector storage.
NOT USED - ALTERNATIVE	pgvector	P3	Alternative Postgres-backed vector storage.
NOT USED - ALTERNATIVE	Milvus	P3	Alternative vector storage for larger deployments.

Reference table continued

Usage / decision	Module	Priority	Purpose or evidence
NOT USED - ALTERNATIVE	Weaviate	P3	Alternative vector database.
USED	bge-reranker	P1	BAAI/bge-reranker-base in existing RAG retrieval.
NOT USED - OPTIONAL	RAGAS	P2	Possible addition to existing custom evaluation.
NOT USED - OPTIONAL	DeepEval	P2	Evaluation alternative; choose based on measured gaps.
NOT USED - OPTIONAL	promptfoo	P2	Prompt/provider regression evaluation option.

Integration update - 17 October 2026

Y: bounded CSV/XLSX conversation tools and scoped lexical-memory tool integration are implemented and covered by the 129-test offline suite. This supersedes the earlier not-connected finding for the shared conversation path. P: broader semantic retrieval, complex-table processing and live acceptance remain. Details: table-memory-tools.md (table-memory-tools.md).